

Chapter 5 Homework: Two-Dimensional Plots

Instructions:

- Write a script file for the exercises below.
- Make sure to write comments in your code and separate output display for clarity.
- Submit your script file in the following format: FirstName_LastInitial_ch5hw.m
Ex: Ryan Cao would submit his lab as: **ryan_c_ch5hw.m**

1. Graph the following functions with the specified command:

a) Use plot: $f(x) = \frac{x^2 - 3x + 7}{\sqrt{2x + 5}}$ for $-1 \leq x \leq 5$

b) Use fplot: $f(x) = (\sin 2x + \cos^2 5x)e^{-0.2x}$ for $-6 \leq x \leq 6$

2. Make a plot of a circle with center (4.2, 2.7) and a radius of 7.5.

3.

The position x as a function of time of a particle that moves along a straight line is given by:

$$x(t) = (-3 + 4t)e^{-0.4t} \text{ ft}$$

The velocity $v(t)$ of the particle is determined by the derivative of $x(t)$ with respect to t , and the acceleration $a(t)$ is determined by the derivative of $v(t)$ with respect to t .

Derive the expressions for the velocity and acceleration of the particle, and make plots of the position, velocity, and acceleration as functions of time for $0 \leq t \leq 20$ s. Use the `subplot` command to make the three plots on the same page with the plot of the position on the top, the velocity in the middle, and the acceleration at the bottom. Label the axes appropriately with the correct units.

4.

The ideal gas law relates the pressure P , volume V , and temperature T of an ideal gas:

$$PV = nRT$$

where n is the number of moles and $R = 8.3145 \text{ J/(K mol)}$. Plots of pressure versus volume at constant temperature are called *isotherms*. Plot the isotherms for one mole of an ideal gas for volume ranging from 1 to 10 m^3 , at temperatures of $T = 100, 200, 300$, and 400 K (four curves in one plot). Label the axes and display a legend. The units for pressure are Pa.

5.

The following data gives the approximate population of the world for selected years from 1850 until 2000.

Year	1850	1910	1950	1980	2000	2010
Population (billions)	1.3	1.75	3	4.4	6	6.8

The population, P , since 1900 can be modeled by the logistic function:

$$P = \frac{11.55}{1 + 18.7e^{-0.0193t}}$$

where P is in billions and t is years since 1850. Make a plot of population versus years. The figure should show the information from the table above as data points and the population modeled by the equation as a solid line. Set the range of the horizontal axis from 1800 to 2200. Add a legend, and label the axes.

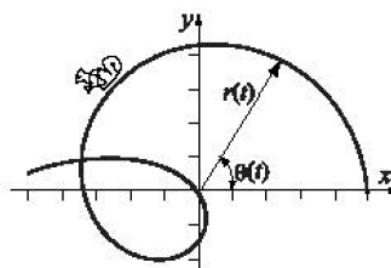
6.

The position as a function of time of a squirrel running on a grass field is given in polar coordinates by:

$$r(t) = 25 + 30[1 - e^{\sin(0.07t)}] \text{ m}$$

$$\theta(t) = 2\pi(1 - e^{-0.2t})$$

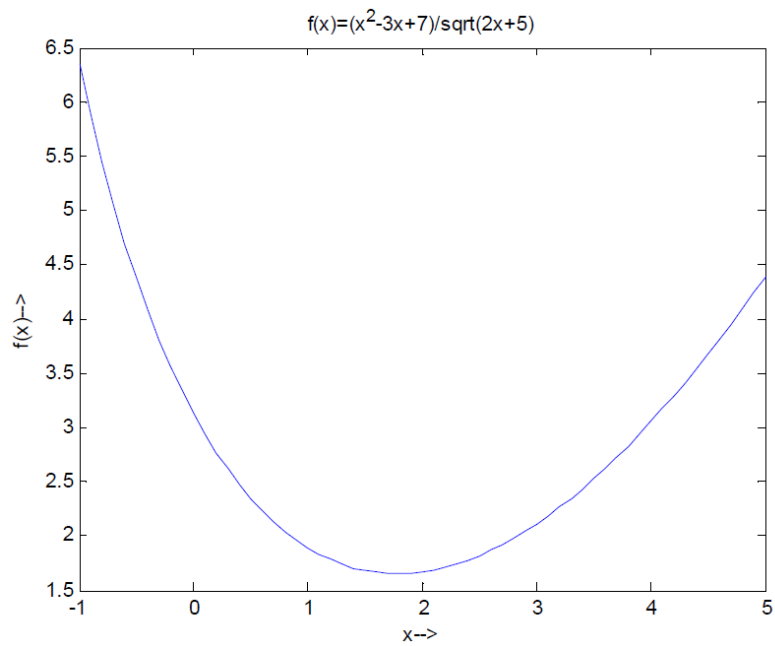
Plot the trajectory (position) of the squirrel for $0 \leq t \leq 20 \text{ s}$.



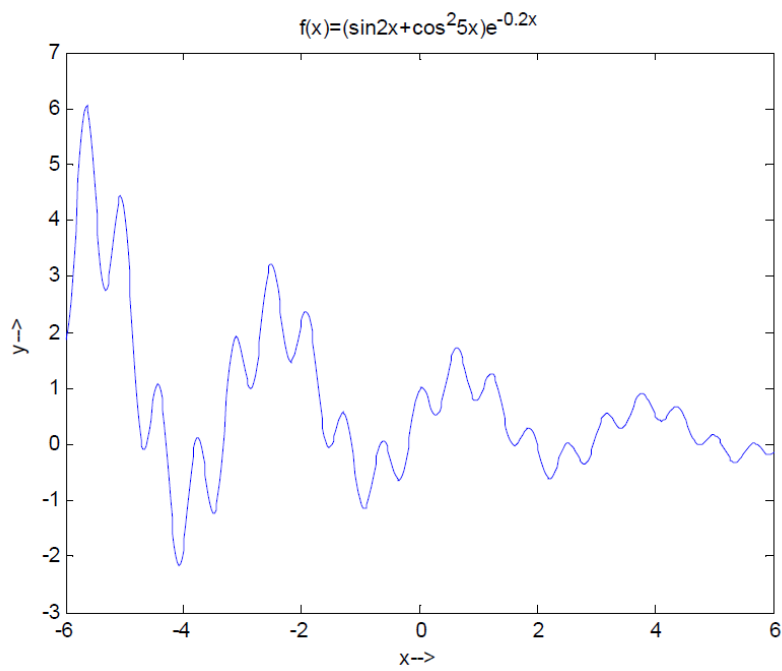
Outputs:

Problem 1:

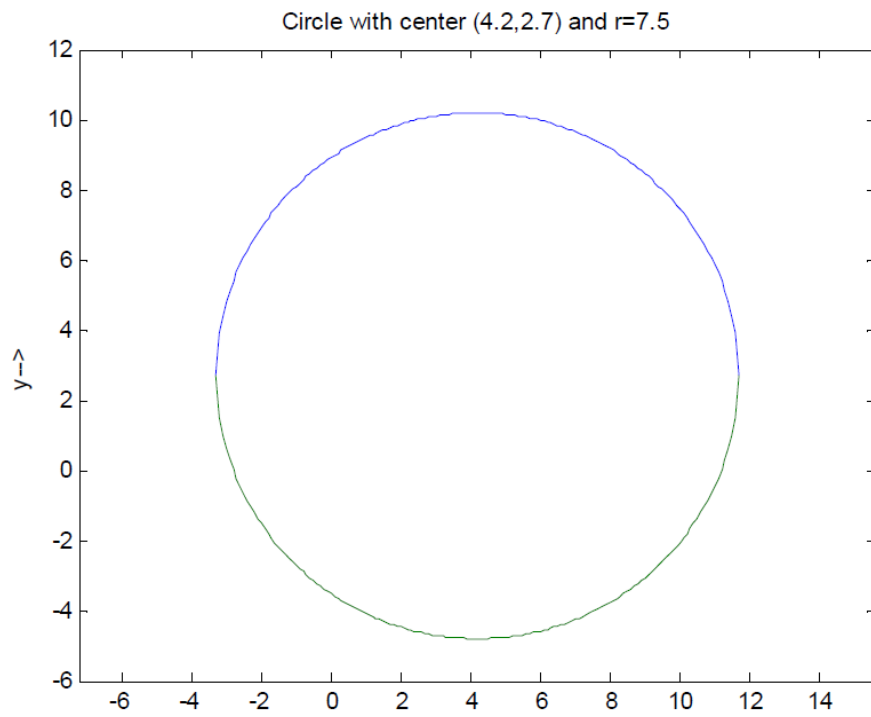
a)



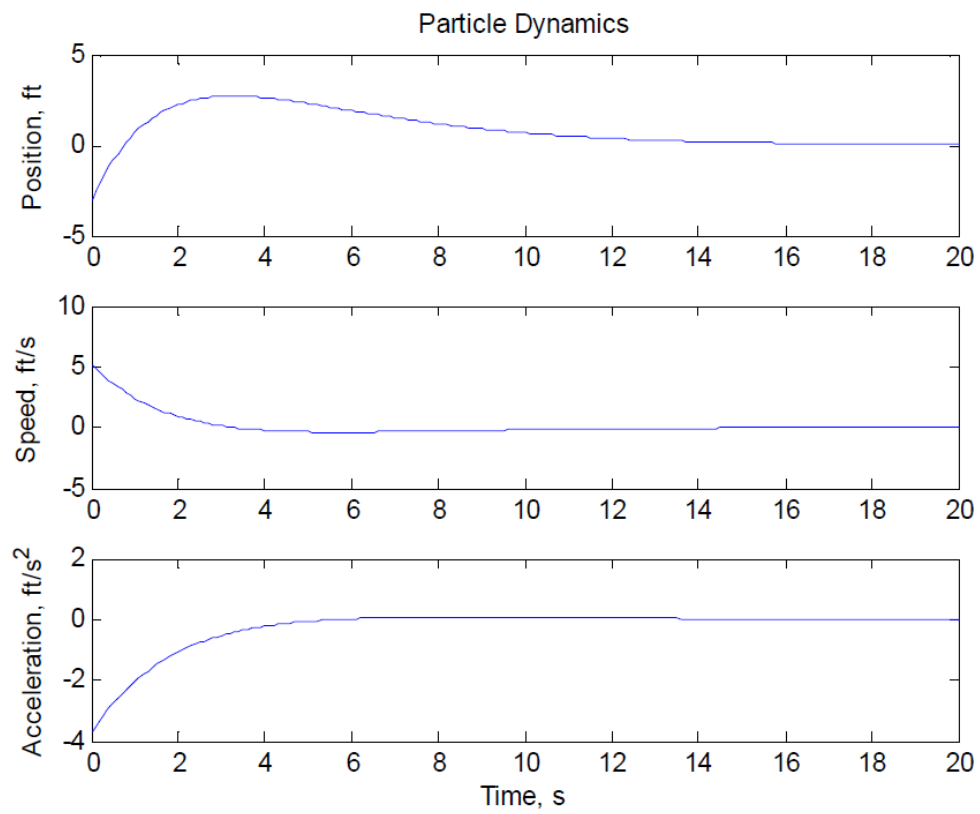
b)



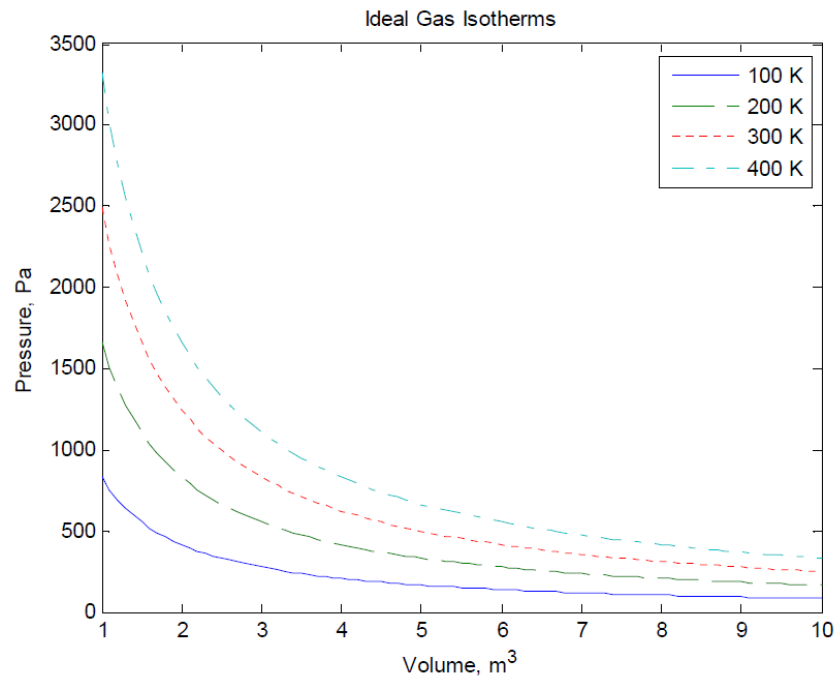
Problem 2:



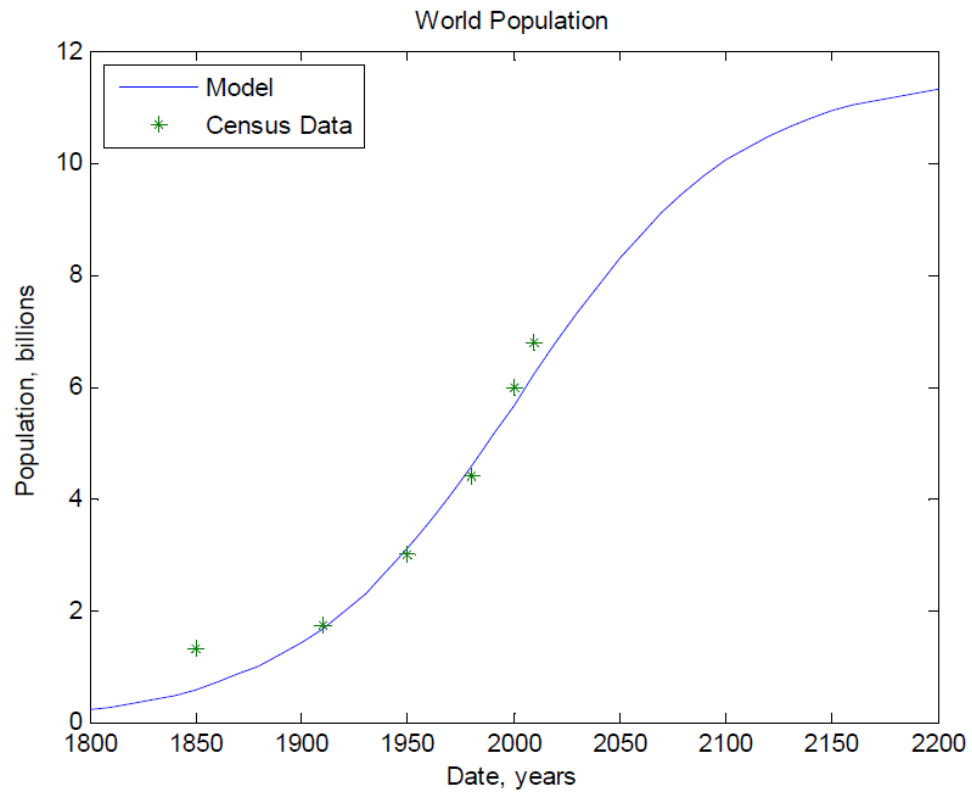
Problem 3:



Problem 4:



Problem 5:



Problem 6:

